1️⃣ Manipulate numbers, graphical and statistical data (Item 7)

Relevance: ★★★★★ (Most critical)

Description:

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

How it appears in this problem:

The task requires:

- Numerical operations
- Ratio calculations
- Unit conversion (hours → minutes)

Specifically:

$$time = \frac{distance}{speed}$$

Students must manipulate the numerical relationship among **6, 18, and 1/3**.

Therefore, this is the **core competency** involved in the problem.

2️⃣ Perform a simple calculation (Item 1)

Relevance: ★★★★★

Description:

> Perform a simple calculation.

Manifestation:

The solution mainly requires a simple computation:

$$6 \div 18$$

followed by a unit conversion.

This represents a **typical basic arithmetic task**.

3️⃣ Understand and utilise constructs based on definitions, rules and formal systems (Item 5)

Relevance: ★★★★☆

Description:

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

Manifestation:

Students must understand the **definition of average speed**:

$$speed = \frac{distance}{time}$$

and rearrange the formula:

$$time = \frac{distance}{speed}$$

This demonstrates the ability to **apply mathematical rules and formulas**.

4️⃣ Use a multi-step procedure leading to a mathematical solution (Item 12)

Relevance: ★★★★☆

Description:

> Use a multi-step procedure leading to a mathematical solution.

Solution steps:

1. Read and interpret the contextual information
2. Identify the average speed formula
3. Substitute the values into the formula
4. Convert hours to minutes
5. Match the result with the correct option

Thus, the task involves a **multi-step problem-solving process**.

5️⃣ Use an understanding of the context to guide solving (Item 13)

Relevance: ★★★☆☆

Description:

> Use an understanding of the context to guide solving.

Manifestation:

Students must understand the real-world meaning of:

- A bicycle trip
- Average speed
- Distance traveled

This contextual understanding helps students construct the mathematical model.

6️⃣ Make sense of and use representations (Item 10)

Relevance: ★★☆☆☆

Description:

> Make sense of, relate and use a variety of representations.

Manifestation:

The problem provides a **verbal description**, which students must convert into:

Text → Mathematical formula

1️⃣ Manipulate numbers, graphical and statistical data（第7项）

相关性：★★★★★（最核心）

能力描述：

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

体现：

本题需要进行：

- 数值运算
- 比例计算
- 单位转换（小时 → 分钟）

具体包括：

$$time = \frac{distance}{speed}$$

学生需要操作 **6、18、1/3 等数值关系。**

因此这是本题最核心能力。

2️⃣ Perform a simple calculation（第1项）

相关性：★★★★★

能力描述：

> Perform a simple calculation.

体现：

解题只需要一个简单计算：

$$6 \div 18$$

然后进行单位换算。

属于典型 **基础运算任务。**

5️⃣ Use an understanding of the context to guide solving（第13项）

相关性：★★★☆☆

能力描述：

> Use an understanding of the context to guide solving.

体现：

学生需要理解：

- 自行车行程
- 平均速度
- 路程

这些现实背景帮助学生建立数学模型。

6️⃣ Make sense of and use representations（第10项）

相关性：★★☆☆☆

能力描述：

> Make sense of, relate and use a variety of representations.

体现：

题目给出的是 **文字情境**，学生需要转换为：

文字 → 数学公式

3️⃣ Understand and utilise constructs based on definitions, rules and formal systems（第5项）

相关性：★★★★☆

能力描述：

> Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms.

体现：

学生必须理解 **平均速度定义**：

$$speed = \frac{distance}{time}$$

并进行 **公式变形**：

$$time = \frac{distance}{speed}$$

这体现了对 **数学规则和公式的应用能力。**

4️⃣ Use a multi-step procedure leading to a mathematical solution（第12项）

相关性：★★★★☆

能力描述：

> Use a multi-step procedure leading to a mathematical solution.

解题步骤：

1️⃣ 阅读情境信息
2️⃣ 识别平均速度公式
3️⃣ 代入数据计算
4️⃣ 将小时转换为分钟
5️⃣ 对应选项

因此属于 **多步骤求解过程。**